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Subject : DSA

Unit 2, S3, SLO1 - Advantages and Disadvantages of an Array

1. Arrays allow ___ access to elements using an index.

Answer: random (direct)

2. Arrays store elements in ___ memory locations.

Answer: contiguous

3. Arrays are useful for storing a ___ number of similar data types.

Answer: fixed (known)

4. The size of an array is ___ once declared.

Answer: fixed (static)

5. Arrays make it easy to perform ___ operations using loops.

Answer: iterative (repetitive)

6. Arrays reduce the ___ of declaring multiple variables.

Answer: overhead (complexity)

7. Memory usage is efficient if the array size is ___.

Answer: known in advance

8. Arrays provide better performance in terms of ___ lookup time.

Answer: $O(1)$ (constant)

9. The size of an array must be known at ___ time.

Answer: compile

10. Arrays have a ___ size, which cannot be changed at runtime.

Answer: static (fixed)

11. Inserting an element in the middle of an array requires ___ other elements.

Answer: shifting

12. Deleting an element in an array also involves ___ elements.

Answer: shifting

13. Wastage of memory can occur if the declared array size is ___.

Answer: overestimated (too large)

14. Arrays can only store data of the ___ type.

Answer: same (homogeneous)

15. Searching in an unsorted array takes ___ time in the worst case.

Answer: $O(n)$ (linear)